HYDROXYCHLOROQUINE

Retinal toxicity — eye screening

Safe in pregnancy

CHEMICAL NAME (RS)-2-[{4-[(7-chloroquinolin-4-yl)amino]pentyl}(ethyl)amino]ethanol (sulfate)

CLASS Antimalarial DMARD (4-aminoquinoline)

TRADE NAMES Plaquenil · Quensyl · Hydroquin · HCQS

THERAPEUTIC USES / INDICATIONS

- Systemic lupus erythematosus (first-line; ↓ flares, ↑ survival) and discoid lupus.
- Rheumatoid arthritis (mild, or combination therapy).
- Sjögren's syndrome; antiphospholipid syndrome (adjunct).
- Malaria prophylaxis/treatment (where chloroquine-sensitive).

ADMINISTRATION

- **PO** with food or milk (↓ GI upset).
- Separate from antacids by 4 h.

DOSAGES

- **SLE / RA:** 200–400 mg daily; max 5 mg/kg/day actual body weight (retinal safety).
- **Malaria prophylaxis:** 400 mg weekly.
- Reduce in renal impairment.

PHARMACOKINETICS

- **Onset:** 6–12 weeks (full effect up to 6 months).
- Large volume of distribution (concentrates in melanin-containing tissues — retina, skin).
- Partly metabolised; excreted in urine. **t½:** 40–50 days.

PHARMACODYNAMICS

- Accumulates in lysosomes → ↑ pH → interferes with **antigen processing** and **Toll-like receptor** signalling → immunomodulation.
- Antiplatelet and lipid/glucose-lowering effects.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, diarrhoea, abdominal cramps.
- Rash, pruritus; skin hyperpigmentation.
- Headache.

ADVERSE EFFECTS (SERIOUS)

- Retinopathy (bull's-eye maculopathy; irreversible) with long-term/high dose.
- Cardiomyopathy, QT prolongation.
- Hypoglycaemia; haemolysis (G6PD).
- Neuromyopathy; psoriasis exacerbation; SJS (rare).

CONTRAINDICATIONS & CAUTIONS

- Pre-existing maculopathy.
- Hypersensitivity to 4-aminoquinolines.
- **Caution:** QT prolongation, G6PD deficiency, psoriasis, porphyria, renal/hepatic impairment.

KEY INTERACTIONS

- **QT-prolonging drugs (azithromycin, antipsychotics):** additive.
- **Antidiabetics, insulin:** hypoglycaemia.
- **Digoxin:** ↑ levels.
- **Antacids:** ↓ absorption.

NURSING MANAGEMENT

- Baseline eye examination within 1 year and annual screening after 5 years (sooner if risk factors).
- Calculate dose by weight.
- Monitor glucose in diabetics; ECG if cardiac risk.
- Assess disease activity.

PATIENT EDUCATION

- Take with food; benefit takes months.
- Attend eye checks; report blurred vision, difficulty reading or seeing colours.
- Keep out of reach of children — small amounts are fatal.
- Safe to continue in pregnancy (lupus).

KEY POINTS

- Antimalarial DMARD; first choice in lupus.
- Retinal toxicity — annual screening after 5 years.
- Max 5 mg/kg/day.
- QT prolongation; hypoglycaemia.